KEVIN SAJAN

kevsaj@gmail.com | Mississauga, ON | kisstudios.net/about | linkedin.com/in/kevin-sajan | github.com/kevsaj

Summary: Software developer with 5+ years building production software for heavy industrial equipment and the web. I write C++/Qt and Python for the operator screens that run underground mining machines, and TypeScript, React and Node for web and mobile apps, including two products I designed and shipped on my own. Release lead on a team that grew from 4 to 12 developers. Calm under pressure, and completing a Master of Data Science (2027).

Technical Skills

- **Languages:** Python, C++, TypeScript, JavaScript, SQL, Java, R
- **Front End & UI:** React, Next.js, React Native, Qt/QML, Angular, Vue, Tailwind, Tkinter
- **Back End & Data:** Node.js, Express, REST APIs, PostgreSQL, MongoDB, OPC UA, Protobuf, ETL, Pandas
- **Cloud & DevOps:** Azure (DevOps pipelines, Kubernetes), Docker, Linux, CI/CD, Git, CMake, Vercel, Firebase
- **Testing:** Jest, Cypress, Qt Test, Puppeteer, Selenium, hardware-in-the-loop test plans
- **Ways of Working:** agile delivery, release management, cross-team collaboration, mentoring, documentation

Work Experience

- **MacLean Engineering** --- Software Developer & Release Lead (Full-time) 2021 July – Current
  - Develop operator interfaces (HMIs) for 5 underground mining machine programs, including a battery-electric grader, in C++/Qt/QML on embedded Linux displays and Python on rugged tablets
  - Built the data layer that streams live machine signals from the controller to the display over OPC UA, with type and status validation so operators never see stale or invalid values
  - Diagnosed a display crash caused by null values in the live-data subscription, fixed it, and added a regression test that reproduces the fault against a simulated server
  - Designed battery status and health screens for an electric machine, tracing data-precision issues to their source and working with controls engineers to correct them
  - Maintain Python remote-control software showing live camera feeds, machine status and fault codes in English and Spanish, delivered as Docker-based installers through Azure Pipelines
  - Built full-stack tools for machine telemetry and reporting with React, Node.js, Express and PostgreSQL, deployed on Azure Kubernetes with Jest and Cypress test suites
  - Built Python and SQL data pipelines that turn raw telemetry into clean, analysis-ready datasets, cutting manual cleanup time by 30–40%
  - Created a 3D point-cloud analysis tool in Python and Open3D with automated clustering and statistical reporting
  - As Release Lead, coordinate deployments across multiple applications, write release test plans with QA and controls engineers, run them on hardware simulation boards, and mentor newer developers
- **Deloitte** --- Financial Advisory & Tax Technician (Full-time) 2019 February – 2021 July
  - Built Python and Puppeteer automation to extract, validate and link financial documents, reducing manual effort and improving accuracy
  - Analyzed invoices, tax returns and financial records to identify discrepancies and unclaimed credits; supported compliance reviews

Projects

- **LoopBot** | kisstudios.net/loopbot: Coding puzzle game for web and Android with 50 levels that teach loops, functions and logic. React, TypeScript, Phaser, Capacitor. Designed, built and published solo.
- **Goodsprout** | kisstudios.net/goodsprout: Private, offline-first journaling app for Android with an encrypted local database and a 3D plant that grows with use. React Native, Expo, SQLCipher, three.js.

Education

University of Pittsburgh (MDS) – Master of Data Science; Data Modeling and AI in Data Management 2025–2027
University of Toronto SCS – Full Stack Web Development Certificate 2020–2021
University of Toronto (HBSc.) – Bachelor of Science; Specialized in Biological and Environmental Sciences 2014–2018

Certifications

- Microsoft AZ-104 Azure Administrator • Microsoft MS-102 Microsoft 365 Administrator • Power BI Data Analyst (CPE) • PMP (in progress)